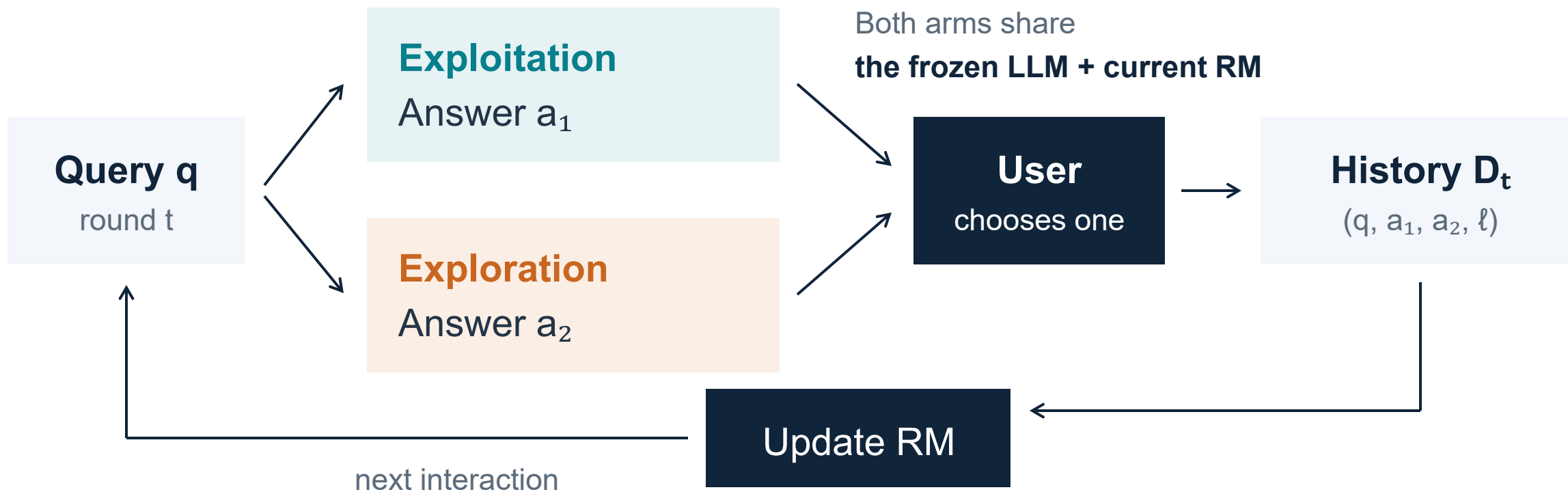
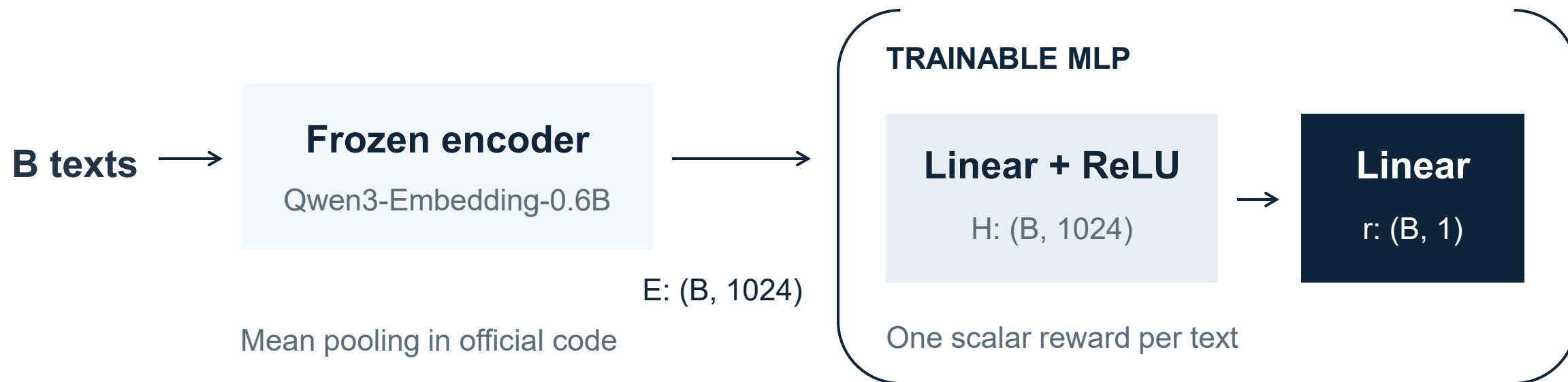


T-POP: one RM, two policies



Two **policies** generate the pair. One **RM** learns from it.

Text $\rightarrow$ vector $\rightarrow$ personal reward



$$H = \text{ReLU}(EW_1 + b_1), \quad \mathbf{r} = HW_2 + b_2$$

$$W_1 \in \mathbb{R}^{1024 \times 1024}, \quad W_2 \in \mathbb{R}^{1024 \times 1}$$

Train on complete response pairs



$$p = \sigma(r_\theta(x_1) - r_\theta(x_2))$$

p = probability that a_1 wins
 $\ell = 1$ if the user prefers a_1

$$\mathcal{L}_t(\theta) = - \sum_{i=1}^t [\ell_i \log p_i + (1 - \ell_i) \log(1 - p_i)] + \lambda \|\theta\|_2^2$$

After feedback: train on all observed pairs. LLM and encoder stay frozen.

RM reward changes the next token

$$\mathbf{s} = \boldsymbol{\pi} + \omega \mathbf{r}, \quad \boldsymbol{\pi}, \mathbf{r}, \mathbf{s} \in \mathbb{R}^C$$

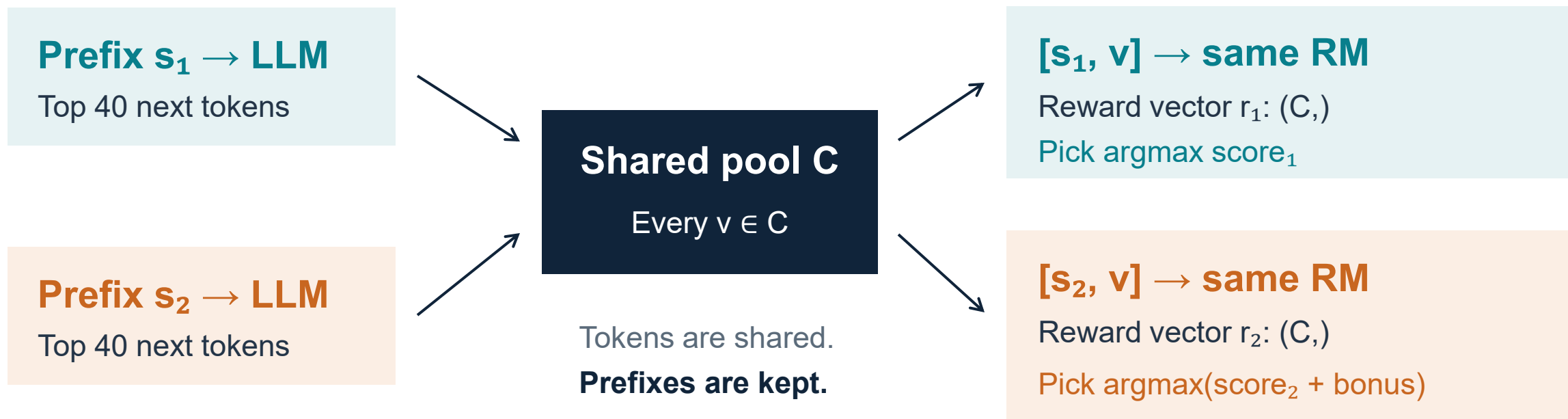
Prefix s: [question; ... Overfitting means] Illustrative values · ω = 1

Token	LLM probability	RM reward	Combined	Pick
memorizing	0.42	0.10	0.52	—
learning	0.35	0.02	0.37	—
fitting	0.18	0.45	0.63	SELECT

RM input: [s, candidate token]. Append the winner and repeat.

Shared candidates, separate prefixes

$$\mathcal{C} = \text{Top40}(\pi_1) \cup \text{Top40}(\pi_2), \quad |\mathcal{C}| \leq 80$$



Exploration also uses the **selected exploitation continuation** as its reference.

Explore uncertain preference differences

EXPLOITATION WINNER (FIXED)

$$g_1^* = \nabla_{\theta} r_{\theta}([s_1, v_1^*])$$

EXPLORATION CANDIDATE v

$$g_2(v) = \nabla_{\theta} r_{\theta}([s_2, v])$$

$$z(v) = g_2(v) - g_1^* \in \mathbb{R}^P, \quad u(v) = \sqrt{z(v)^{\top} V^{-1} z(v)}$$

EXPLORATION SELECTION

$$v_2^* = \arg \max_{v \in \mathcal{C}} [\pi_2(v) + \omega r_{\theta}([s_2, v]) + \omega v u(v)]$$

P = RM parameter count. V tracks covered gradient-difference directions.

Two clocks: update V, then train RM

Every token

Algorithm 1 · line 14

$$\delta = g_1^* - g_2^*, \quad V \leftarrow V + \delta \delta^\top, \quad V \in \mathbb{R}^{P \times P}$$

$p = 1, 2, 3, \dots$

Selected prefixes · θ fixed · No preference label needed



Both responses finish; the user chooses one.

After feedback

Algorithm 1 · line 17

$$\theta_{t+1} = \arg \min_{\theta} \mathcal{L}_t(\theta)$$

Complete response pair → add to D_t → train the shared RM

Deployment returns one answer

ONLINE PERSONALIZATION

2

answers per query

Pairwise feedback

RM keeps learning

FINAL DEPLOYMENT

1

answer per query

Exploitation only

Learned RM is frozen

One RM. Two policies. Two clocks.

Evidence: larger Mistral/Qwen gains; near tie with AMULET on Llama.